Brutal Practice Paper B20 - Reproduction in Plants | The Triangle & its Properties | Light | Symmetry

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. A flower has 5 stamens. Each anther carries 4 pollen sacs, and each sac holds 220 pollen grains. How many pollen grains can this one flower release in all? [Reproduction in Plants]

- **(A)** 8800 grains
- **(B)** 4400 grains
- **(C)** 2200 grains
- **(D)** 1100 grains
- **(E)** 440 grains
- **(F)** 880 grains

2. An exterior angle of a triangle measures 125 deg, and one of its two remote interior angles is 53 deg. Find the third angle of the triangle - the one that sits next to this exterior angle. [The Triangle & its Properties]

- **(A)** 53 deg
- **(B)** 65 deg
- **(C)** 55 deg
- **(D)** 35 deg
- **(E)** 108 deg
- **(F)** 72 deg

3. Two plane mirrors are placed facing each other at an angle of 60 deg. How many images of a small object placed between them are formed? [Light]

- **(A)** 6 images
- **(B)** 3 images
- **(C)** 7 images
- **(D)** 60 images
- **(E)** 5 images
- **(F)** 4 images

4. A regular hexagon is turned about its centre until it looks exactly the same as before. What is the SMALLEST angle of rotation that does this? [Symmetry]

- **(A)** 120 deg
- **(B)** 90 deg
- **(C)** 60 deg
- **(D)** 30 deg
- **(E)** 6 deg
- **(F)** 72 deg
- **(G)** 45 deg

5. A nursery sows 250 seeds. Records show 12% of them never sprout. How many seedlings actually come up? [Reproduction in Plants]

- **(A)** 88 seedlings
- **(B)** 12 seedlings
- **(C)** 220 seedlings
- **(D)** 200 seedlings
- **(E)** 238 seedlings
- **(F)** 262 seedlings
- **(G)** 30 seedlings

6. In a triangle the three angles are split in the ratio 2 : 3 : 4. Find the biggest angle. [The Triangle & its Properties]

- **(A)** 70 deg
- **(B)** 60 deg
- **(C)** 100 deg
- **(D)** 90 deg
- **(E)** 45 deg
- **(F)** 80 deg

7. Two plane mirrors stand at right angles (90 deg) to each other. How many images of an object between them are seen? [Light]

- **(A)** 4 images
- **(B)** 2 images
- **(C)** 1 image
- **(D)** 6 images
- **(E)** 3 images
- **(F)** 5 images

8. An equilateral triangle has rotational symmetry. What is its ORDER of rotational symmetry, and the smallest angle of rotation? [Symmetry]

- **(A)** order 3, angle 120 deg
- **(B)** order 6, angle 60 deg
- **(C)** order 1, angle 360 deg
- **(D)** order 2, angle 180 deg
- **(E)** order 4, angle 90 deg
- **(F)** order 3, angle 60 deg

9. New little plants grow straight out of the buds sitting on the notched edges of a Bryophyllum leaf, with no seed involved. This way of making new plants is called: [Reproduction in Plants]

- **(A)** spore formation
- **(B)** sexual reproduction
- **(C)** budding, as in yeast
- **(D)** vegetative propagation by leaves
- **(E)** fertilisation
- **(F)** fragmentation

10. An isosceles triangle has a vertex (apex) angle of 40 deg. What is the measure of an exterior angle at one of its base vertices? [The Triangle & its Properties]

- **(A)** 20 deg
- **(B)** 100 deg
- **(C)** 70 deg
- **(D)** 110 deg
- **(E)** 140 deg
- **(F)** 160 deg

11. A ray of light strikes a plane mirror, making an angle of 35 deg with the MIRROR SURFACE. What is the angle between the incident ray and the reflected ray? [Light]

- **(A)** 35 deg
- **(B)** 55 deg
- **(C)** 90 deg
- **(D)** 70 deg
- **(E)** 110 deg
- **(F)** 125 deg
- **(G)** 140 deg

12. How many lines of symmetry does a square have, and what is its order of rotational symmetry? [Symmetry]

- **(A)** 2 lines, order 4
- **(B)** 4 lines, order 8
- **(C)** 1 line, order 4
- **(D)** 4 lines of symmetry, rotational order 4
- **(E)** 2 lines, order 2
- **(F)** 8 lines, order 4

13. One potato tuber has 8 eyes; each eye can grow into a plant, and each plant produces 6 new tubers, every new tuber again having 8 eyes. How many eyes are there altogether on the new tubers? [Reproduction in Plants]

- **(A)** 64 eyes
- **(B)** 96 eyes
- **(C)** 48 eyes
- **(D)** 8 eyes
- **(E)** 384 eyes
- **(F)** 56 eyes
- **(G)** 240 eyes

14. Two sides of a triangle are 7 cm and 10 cm. How many different WHOLE-NUMBER lengths are possible for the third side? [The Triangle & its Properties]

- **(A)** 14 values
- **(B)** 13 values
- **(C)** 12 values
- **(D)** 17 values
- **(E)** 16 values
- **(F)** 15 values

15. An object stands 8 cm in front of a plane mirror. It is then pushed 3 cm TOWARDS the mirror. After moving, what is the distance between the object and its image? [Light]

- **(A)** 16 cm
- **(B)** 6 cm
- **(C)** 13 cm
- **(D)** 8 cm
- **(E)** 5 cm
- **(F)** 10 cm

16. A rectangle that is NOT a square is examined. How many lines of symmetry does it have, and what is its rotational order? [Symmetry]

- **(A)** 2 lines of symmetry, rotational order 2
- **(B)** 1 line, order 2
- **(C)** 4 lines, order 2
- **(D)** 4 lines, order 4
- **(E)** 2 lines, order 4
- **(F)** 0 lines, order 2
- **(G)** 2 lines, order 1

17. A flower is pollinated by wind. Which set of features is it most likely to have? [Reproduction in Plants]

- **(A)** brightly coloured petals to attract bees
- **(B)** heavy sticky pollen and a smooth stigma
- **(C)** large nectar glands and broad sticky stigma
- **(D)** small dull flowers with light, dry pollen and feathery stigmas
- **(E)** scented flowers with a few heavy pollen grains
- **(F)** large colourful flowers full of nectar
- **(G)** bright petals, sweet scent and sticky pollen

18. A right-angled triangle has legs of 9 cm and 12 cm. Find the length of the hypotenuse. [The Triangle & its Properties]

- **(A)** 105 cm
- **(B)** 11 cm
- **(C)** 225 cm
- **(D)** 21 cm
- **(E)** 15 cm
- **(F)** 18 cm

19. A wall clock seen in a plane mirror appears to show 9:00. What is the ACTUAL time on the clock? [Light]

- **(A)** 2:00
- **(B)** 12:00
- **(C)** 6:00
- **(D)** 9:00
- **(E)** 3:00
- **(F)** 8:00
- **(G)** 3:30

20. A regular pentagon is rotated about its centre. What is the smallest angle of rotation that maps it onto itself? [Symmetry]

- **(A)** 72 deg
- **(B)** 36 deg
- **(C)** 5 deg
- **(D)** 120 deg
- **(E)** 60 deg
- **(F)** 90 deg
- **(G)** 108 deg

21. Looking at four flowers, you must pick the one that is bisexual. A bisexual flower is one that has: [Reproduction in Plants]

- **(A)** only an ovary and petals
- **(B)** two pistils but no stamens
- **(C)** neither stamens nor pistil
- **(D)** only a pistil
- **(E)** only stamens
- **(F)** petals but no reproductive parts
- **(G)** both stamens and pistil in the same flower

22. A straight ladder 13 m long leans against a wall, with its foot 5 m out from the base of the wall. How high up the wall does the top of the ladder reach? [The Triangle & its Properties]

- **(A)** 18 m
- **(B)** 9 m
- **(C)** 144 m
- **(D)** 8 m
- **(E)** 12 m
- **(F)** 11 m

23. Which type of mirror is used as the side and rear-view mirror in cars and scooters, because it gives a wide view of the traffic behind? [Light]

- **(A)** plane mirror only
- **(B)** a flat glass sheet
- **(C)** a prism
- **(D)** a converging lens
- **(E)** concave mirror
- **(F)** convex mirror

24. The capital letter H is studied for symmetry. How many lines of symmetry does it have, and what is its rotational order? [Symmetry]

- **(A)** 2 lines, order 4
- **(B)** 2 lines, order 1
- **(C)** 1 line, order 1
- **(D)** 1 line, order 2
- **(E)** 4 lines, order 4
- **(F)** 0 lines, order 2
- **(G)** 2 lines of symmetry, rotational order 2

25. A fern frond bears 3 spore cases, each holding 64 spores. On average only 1 spore in every 16 lands well and grows into a new fern. How many new ferns form? [Reproduction in Plants]

- **(A)** 48 ferns
- **(B)** 3 ferns
- **(C)** 64 ferns
- **(D)** 4 ferns
- **(E)** 16 ferns
- **(F)** 12 ferns

26. Which line in a triangle joins a vertex to the MIDPOINT of the opposite side? [The Triangle & its Properties]

- **(A)** the base
- **(B)** a perpendicular bisector of a side
- **(C)** a median
- **(D)** an angle bisector
- **(E)** an altitude
- **(F)** the hypotenuse

27. A dentist wants to see an enlarged, upright image of a tooth held close to the mirror. Which mirror should be used? [Light]

- **(A)** a diverging lens
- **(B)** convex mirror
- **(C)** plane mirror
- **(D)** concave mirror
- **(E)** a prism
- **(F)** a glass slab

28. A regular octagon (like a STOP sign) is rotated about its centre. What is the smallest angle of rotation that brings it onto itself? [Symmetry]

- **(A)** 135 deg
- **(B)** 30 deg
- **(C)** 90 deg
- **(D)** 8 deg
- **(E)** 45 deg
- **(F)** 40 deg
- **(G)** 60 deg

29. Which seed is best built for being carried away by the wind? [Reproduction in Plants]

- **(A)** a heavy coconut with a thick husk
- **(B)** a seed inside a pod that bursts open
- **(C)** a juicy berry eaten by birds
- **(D)** a seed with tiny hooks that stick to fur
- **(E)** a seed coated in sticky gum
- **(F)** a large dense seed that sinks in water
- **(G)** a light seed with wings or a tuft of hair

30. In triangle ABC, angle A = (x) deg, angle B = (x + 20) deg and angle C = (x + 40) deg. Find the SMALLEST angle. [The Triangle & its Properties]

- **(A)** 45 deg
- **(B)** 60 deg
- **(C)** 20 deg
- **(D)** 80 deg
- **(E)** 100 deg
- **(F)** 40 deg
- **(G)** 50 deg

31. When white light passes through a prism it splits into a band of seven colours. Which colour bends (deviates) the MOST? [Light]

- **(A)** red
- **(B)** yellow
- **(C)** green
- **(D)** blue
- **(E)** indigo
- **(F)** orange
- **(G)** violet

32. A parallelogram that is not a rectangle or rhombus is examined. How many lines of symmetry does it have, and what is its rotational order? [Symmetry]

- **(A)** 0 lines, order 1
- **(B)** 1 line, order 2
- **(C)** 2 lines, order 2
- **(D)** 0 lines of symmetry, rotational order 2
- **(E)** 2 lines, order 1
- **(F)** 4 lines, order 4
- **(G)** 1 line, order 1

33. A maize cob has 16 rows of kernels with 30 kernels in each row. After harvest, 5% of the kernels turn out to be empty. How many kernels are filled with grain? [Reproduction in Plants]

- **(A)** 480 kernels
- **(B)** 24 kernels
- **(C)** 476 kernels
- **(D)** 456 kernels
- **(E)** 455 kernels
- **(F)** 450 kernels
- **(G)** 504 kernels

34. An exterior angle of an isosceles triangle is 110 deg, formed at the vertex where the two EQUAL base angles' opposite... the two remote interior angles are equal. What is each of those two equal angles? [The Triangle & its Properties]

(A) 55 deg
(B) 70 deg
(C) 35 deg
(D) 50 deg
(E) 110 deg
(F) 65 deg
(G) 60 deg

35. Which list correctly describes the image formed by a plane mirror? [Light]

(A) virtual, erect, magnified, not inverted
(B) real, erect, diminished
(C) real, erect, same size
(D) virtual, inverted, diminished
(E) virtual, erect, same size, and laterally inverted
(F) virtual, inverted, same size, not inverted

36. How many lines of symmetry does a circle have, and what is its order of rotational symmetry? [Symmetry]

(A) 8 lines, order 8
(B) 1 line, order 1
(C) 0 lines, order 1
(D) 2 lines, order 2
(E) 4 lines, order 4
(F) infinitely many lines, and infinite rotational order
(G) 360 lines, order 360

37. After fertilisation in a flower, which change is correct? [Reproduction in Plants]

(A) the style becomes the seed
(B) the stigma becomes a fruit
(C) the sepals become the fruit
(D) the petals become seeds
(E) the ovary becomes a seed and the ovule becomes a fruit
(F) the anther becomes a fruit
(G) the ovule becomes a seed and the ovary becomes a fruit

38. In a right-angled triangle, one of the acute angles is 35 deg. What is the OTHER acute angle? [The Triangle & its Properties]

(A) 35 deg
(B) 125 deg
(C) 145 deg
(D) 55 deg
(E) 65 deg
(F) 45 deg
(G) 90 deg

39. Two plane mirrors are set at an angle of 72 deg. How many images of an object placed between them are formed? [Light]

- **(A)** 2 images
- **(B)** 6 images
- **(C)** 5 images
- **(D)** 3 images
- **(E)** 7 images
- **(F)** 4 images
- **(G)** 72 images

40. The capital letter S has rotational symmetry. How many lines of symmetry does it have? [Symmetry]

- **(A)** 2 lines of symmetry
- **(B)** infinite lines
- **(C)** 1 line of symmetry
- **(D)** 1 vertical line only
- **(E)** 4 lines of symmetry
- **(F)** 1 horizontal line only
- **(G)** 0 lines of symmetry (but it has rotational order 2)

41. Pollen from a flower lands on the stigma of a DIFFERENT plant of the same kind. This is called: [Reproduction in Plants]

- **(A)** fragmentation
- **(B)** self-pollination
- **(C)** fertilisation
- **(D)** double fertilisation
- **(E)** vegetative propagation
- **(F)** germination
- **(G)** cross-pollination

42. The hypotenuse of a right triangle is 25 cm and one leg is 7 cm. Find the other leg. [The Triangle & its Properties]

- **(A)** 20 cm
- **(B)** 18 cm
- **(C)** 576 cm
- **(D)** 24 cm
- **(E)** 26 cm
- **(F)** 32 cm
- **(G)** 12 cm

43. Two plane mirrors are placed at an angle of 40 deg. How many images are formed? [Light]

- **(A)** 9 images
- **(B)** 40 images
- **(C)** 7 images
- **(D)** 6 images
- **(E)** 8 images
- **(F)** 10 images
- **(G)** 5 images

44. A shape has rotational symmetry of order 9. What is its smallest angle of rotation? [Symmetry]

- **(A)** 40 deg
- **(B)** 20 deg
- **(C)** 9 deg
- **(D)** 30 deg
- **(E)** 36 deg
- **(F)** 45 deg
- **(G)** 90 deg

45. A plant opens 8 flowers; each flower sets a fruit containing 25 seeds. If only 60% of all these seeds sprout, how many new plants can grow? [Reproduction in Plants]

- **(A)** 125 plants
- **(B)** 80 plants
- **(C)** 120 plants
- **(D)** 48 plants
- **(E)** 100 plants
- **(F)** 200 plants
- **(G)** 150 plants

46. Two angles of a triangle are 48 deg and 67 deg. After finding the third angle, how is the triangle classified by its angles? [The Triangle & its Properties]

- **(A)** straight triangle
- **(B)** reflex-angled triangle
- **(C)** acute-angled triangle
- **(D)** equilateral triangle
- **(E)** right-angled triangle
- **(F)** obtuse-angled triangle
- **(G)** isosceles right triangle

47. A ray hits a plane mirror and reflects. If the MIRROR is then turned by 15 deg (the incident ray staying fixed), by how much does the reflected ray turn? [Light]

- **(A)** 0 deg
- **(B)** 15 deg
- **(C)** 90 deg
- **(D)** 30 deg
- **(E)** 45 deg
- **(F)** 7.5 deg
- **(G)** 60 deg

48. A figure looks exactly the same after being rotated by 36 deg about its centre. What is its ORDER of rotational symmetry? [Symmetry]

- **(A)** 12
- **(B)** 36
- **(C)** 6
- **(D)** 9
- **(E)** 5
- **(F)** 8
- **(G)** 10

49. The pistil (the female part of a flower) is made of which three parts, from top to bottom? [Reproduction in Plants]

- **(A)** stigma, anther and ovary
- **(B)** stigma, filament and anther
- **(C)** sepal, petal and ovary
- **(D)** filament, style and stigma
- **(E)** stigma, style and ovary
- **(F)** anther, filament and ovary
- **(G)** anther, style and stigma

50. What do all three exterior angles of any triangle add up to (taking one exterior angle at each vertex)? [The Triangle & its Properties]

- **(A)** 90 deg
- **(B)** 540 deg
- **(C)** 360 deg
- **(D)** 270 deg
- **(E)** 180 deg
- **(F)** 300 deg

51. Of these mirrors, which one can throw a REAL image onto a screen? [Light]

- **(A)** a convex mirror
- **(B)** none of them ever can
- **(C)** a convex mirror held very close
- **(D)** a flat polished metal sheet
- **(E)** a diverging mirror
- **(F)** a concave mirror
- **(G)** a plane mirror

52. A regular polygon has 12 equal sides. What is its smallest angle of rotation, and how many lines of symmetry does it have? [Symmetry]

- **(A)** angle 12 deg, with 12 lines
- **(B)** angle 150 deg, with 12 lines
- **(C)** angle 30 deg, with 12 lines of symmetry
- **(D)** angle 60 deg, with 12 lines
- **(E)** angle 30 deg, with 24 lines
- **(F)** angle 36 deg, with 12 lines

53. Which organism reproduces by breaking into pieces, where each piece grows into a new individual (fragmentation)? [Reproduction in Plants]

- **(A)** a sunflower
- **(B)** a fern
- **(C)** a pea plant
- **(D)** a mango tree
- **(E)** yeast
- **(F)** Spirogyra

54. A right-angled triangle has its two legs equal to 6 cm and 8 cm. What is its AREA? [The Triangle & its Properties]

- **(A)** 14 cm^2
- **(B)** 30 cm^2
- **(C)** 10 cm^2
- **(D)** 48 cm^2
- **(E)** 40 cm^2
- **(F)** 24 cm^2
- **(G)** 20 cm^2

55. In a primary rainbow, which colour appears on the OUTERMOST edge of the arc? [Light]

- **(A)** green
- **(B)** red
- **(C)** violet
- **(D)** indigo
- **(E)** orange
- **(F)** yellow
- **(G)** blue

56. Among the capital letters N, Z and S, what symmetry do they SHARE? [Symmetry]

- **(A)** they each have a vertical line of symmetry
- **(B)** they have infinite lines of symmetry
- **(C)** they have rotational symmetry of order 2 but no line of symmetry
- **(D)** they each have 1 line of symmetry
- **(E)** they have no symmetry at all
- **(F)** they have rotational order 4
- **(G)** they each have 2 lines of symmetry

57. In which organism does a small bulge (a bud) grow on the parent, then break off as a new individual? [Reproduction in Plants]

- **(A)** Spirogyra
- **(B)** a moss plant
- **(C)** a coconut palm
- **(D)** a bread mould
- **(E)** yeast
- **(F)** a rose bush

58. A triangular flag is cut with a base of 14 cm and a height of 9 cm. Find its area. [The Triangle & its Properties]

- **(A)** 63 cm^2
- **(B)** 112 cm^2
- **(C)** 23 cm^2
- **(D)** 252 cm^2
- **(E)** 31.5 cm^2
- **(F)** 46 cm^2
- **(G)** 126 cm^2

59. A simple periscope (used to see over walls or in submarines) works using two plane mirrors. At what angle is each mirror usually fixed to the tube? [Light]

- **(A)** 0 deg
- **(B)** 60 deg
- **(C)** 45 deg
- **(D)** 90 deg
- **(E)** 22.5 deg
- **(F)** 15 deg

60. An isosceles triangle that is NOT equilateral is examined. How many lines of symmetry does it have, and what is its rotational order? [Symmetry]

- **(A)** 0 lines, order 1
- **(B)** 2 lines, order 1
- **(C)** 1 line of symmetry, rotational order 1
- **(D)** 1 line, order 2
- **(E)** 3 lines, order 3
- **(F)** 1 line, order 3
- **(G)** 2 lines, order 2

61. A group of organisms reproduces by making tiny spores that float in air and grow when they land somewhere damp. This group is best represented by: [Reproduction in Plants]

- **(A)** moss, fern and fungus
- **(B)** potato, ginger and onion
- **(C)** sunflower, marigold and lotus
- **(D)** apple, orange and grape
- **(E)** mango, neem and rose
- **(F)** bryophyllum and money-plant

62. In triangle ABC, angle A is twice angle B, and angle C is three times angle B. What is the LARGEST angle? [The Triangle & its Properties]

- **(A)** 45 deg
- **(B)** 75 deg
- **(C)** 60 deg
- **(D)** 120 deg
- **(E)** 30 deg
- **(F)** 100 deg
- **(G)** 90 deg

63. Light from a torch passes through a tiny straight gap and casts a sharp-edged shadow of a card behind it. This shows that: [Light]

- **(A)** light travels in straight lines
- **(B)** shadows form only in water
- **(C)** light has no fixed direction
- **(D)** light is made only of sound
- **(E)** light bends around the card
- **(F)** light travels in curved paths

64. A rhombus (a 'diamond' shape that is not a square) is examined. How many lines of symmetry does it have, and what is its rotational order? [Symmetry]

- **(A)** 2 lines of symmetry, rotational order 2
- **(B)** 2 lines, order 1
- **(C)** 4 lines, order 4
- **(D)** 1 line, order 2
- **(E)** 0 lines, order 2
- **(F)** 2 lines, order 4

65. A packet states a germination rate of 84%. If a farmer sows 500 such seeds, how many are expected to sprout? [Reproduction in Plants]

- **(A)** 420 seeds
- **(B)** 80 seeds
- **(C)** 440 seeds
- **(D)** 416 seeds
- **(E)** 458 seeds
- **(F)** 400 seeds

66. An isosceles triangle has two equal sides of 8 cm each. How many WHOLE-NUMBER lengths are possible for the base? [The Triangle & its Properties]

- **(A)** 14 values
- **(B)** 7 values
- **(C)** 31 values
- **(D)** 17 values
- **(E)** 16 values
- **(F)** 15 values
- **(G)** 8 values

67. Two plane mirrors are placed at an angle of 30 deg. How many images are formed? [Light]

- **(A)** 6 images
- **(B)** 30 images
- **(C)** 10 images
- **(D)** 12 images
- **(E)** 13 images
- **(F)** 11 images

68. A wheel design looks exactly the same after a turn of one-third of a full circle. What is its order of rotational symmetry and the angle of that turn? [Symmetry]

- **(A)** order 3, turn of 120 deg
- **(B)** order 3, turn of 90 deg
- **(C)** order 4, turn of 90 deg
- **(D)** order 2, turn of 180 deg
- **(E)** order 1, turn of 360 deg
- **(F)** order 3, turn of 60 deg
- **(G)** order 6, turn of 120 deg

69. Some water plants are pollinated with the help of water itself. A plant that uses WATER as its pollinating agent is: [Reproduction in Plants]

- **(A)** a dry desert cactus
- **(B)** a self-pollinating pea in air
- **(C)** Hydrilla (a submerged water plant)
- **(D)** a wind-pollinated grass
- **(E)** a bird-pollinated flower
- **(F)** a mango tree pollinated by flies
- **(G)** a bee-pollinated sunflower

70. A rectangular field measures 24 m by 7 m. A straight path runs along its diagonal. How long is the path? [The Triangle & its Properties]

- **(A)** 625 m
- **(B)** 12.5 m
- **(C)** 25 m
- **(D)** 17 m
- **(E)** 31 m
- **(F)** 26 m
- **(G)** 20 m

71. Which statement about a CONVEX mirror's image of a normal object is always true? [Light]

- **(A)** the image is virtual, erect and smaller than the object
- **(B)** the image is inverted and smaller
- **(C)** the image is real and the same size
- **(D)** the image is virtual but always larger
- **(E)** the image can be caught on a screen
- **(F)** the image is real and erect
- **(G)** the image is real, inverted and larger

72. Which capital letter has BOTH a vertical and a horizontal line of symmetry? [Symmetry]

- **(A)** S
- **(B)** T
- **(C)** P
- **(D)** X
- **(E)** V
- **(F)** E
- **(G)** A

73. A pea pod holds 12 seeds. Insects eat one-fourth of them, and then one-third of the seeds that are LEFT rot in the rain. How many healthy seeds remain? [Reproduction in Plants]

- **(A)** 8 seeds
- **(B)** 3 seeds
- **(C)** 7 seeds
- **(D)** 9 seeds
- **(E)** 4 seeds
- **(F)** 6 seeds
- **(G)** 5 seeds

74. The three angles of a triangle are in the ratio 3 : 4 : 5. What is the largest angle, and is the triangle right-angled? [The Triangle & its Properties]

- **(A)** 75 deg, and it is right-angled
- **(B)** 90 deg, and it is right-angled
- **(C)** 60 deg, not right-angled
- **(D)** 75 deg, and it is NOT right-angled
- **(E)** 100 deg, not right-angled
- **(F)** 45 deg, not right-angled

75. A capital letter looks exactly the SAME as the original when seen in a plane mirror (no left-right change shows). Which letter behaves like this? [Light]

- **(A)** L
- **(B)** A
- **(C)** P
- **(D)** F
- **(E)** B
- **(F)** G
- **(G)** R

76. A regular polygon is rotated about its centre and first looks the same after a turn of 24 deg. How many SIDES does the polygon have? [Symmetry]

- **(A)** 15 sides
- **(B)** 8 sides
- **(C)** 12 sides
- **(D)** 6 sides
- **(E)** 24 sides
- **(F)** 10 sides
- **(G)** 20 sides

77. Fertilisation in a flowering plant means: [Reproduction in Plants]

- **(A)** a seed soaks up water and sprouts
- **(B)** the male gamete fuses with the female gamete to form a zygote
- **(C)** the ovary swells into a fruit
- **(D)** pollen is carried by a bee
- **(E)** a bud grows on a parent and breaks off
- **(F)** pollen lands on the stigma
- **(G)** petals fall off after flowering

78. A right-angled triangle has legs of 20 cm and 21 cm. Find the hypotenuse. [The Triangle & its Properties]

- **(A)** 841 cm
- **(B)** 25 cm
- **(C)** 30 cm
- **(D)** 41 cm
- **(E)** 28 cm
- **(F)** 29 cm

79. A pencil standing in a glass of water looks bent at the water's surface. This happens because light: [Light]

- **(A)** changes speed and bends as it passes from water to air (refraction)
- **(B)** reflects off the pencil only
- **(C)** turns into sound underwater
- **(D)** bounces straight back from the water
- **(E)** is completely absorbed by water
- **(F)** stops at the surface of the water

80. A semicircle (half of a circle) is examined for symmetry. How many lines of symmetry does it have, and what is its rotational order? [Symmetry]

- **(A)** 0 lines, order 1
- **(B)** 1 line, order 2
- **(C)** 1 line of symmetry, rotational order 1
- **(D)** 1 line, order 4
- **(E)** 2 lines, order 1
- **(F)** infinite lines, infinite order
- **(G)** 2 lines, order 2

81. A seed needs about 84 days from sowing to harvest. If you sow it today, roughly how many WEEKS until harvest? [Reproduction in Plants]

- **(A)** 8 weeks
- **(B)** 84 weeks
- **(C)** 14 weeks
- **(D)** 12 weeks
- **(E)** 21 weeks
- **(F)** 6 weeks
- **(G)** 10 weeks

82. A triangle is drawn on paper. How many medians can it have in all, and at which single point do they cross? [The Triangle & its Properties]

- **(A)** 3 medians, meeting at the orthocentre
- **(B)** 4 medians, meeting at the centre
- **(C)** 3 medians, meeting at the centroid
- **(D)** 2 medians, meeting at the centroid
- **(E)** 1 median through the right angle
- **(F)** 3 medians that never meet
- **(G)** 3 altitudes, meeting at the centroid

83. The angle of incidence of a light ray on a plane mirror is 0 deg (it hits straight along the normal). What is the angle of reflection? [Light]

- **(A)** 90 deg
- **(B)** 180 deg
- **(C)** 30 deg
- **(D)** 45 deg
- **(E)** 0 deg (it reflects straight back along the same path)
- **(F)** 60 deg
- **(G)** cannot reflect

84. A regular polygon has exactly 9 lines of symmetry. What is its smallest angle of rotation? [Symmetry]

(A) 45 deg
(B) 20 deg
(C) 40 deg
(D) 60 deg
(E) 36 deg
(F) 90 deg

85. In which plant do the ripe fruits suddenly burst open and fling their seeds away (dispersal by bursting)? [Reproduction in Plants]

(A) maple
(B) balsam (and pea/castor pods)
(C) lotus floating on water
(D) cocklebur with hooks
(E) coconut
(F) sunflower carried by birds
(G) dandelion

86. One angle of a triangle is 90 deg. The other two are in the ratio 2 : 3. Find the larger of these two angles. [The Triangle & its Properties]

(A) 45 deg
(B) 54 deg
(C) 90 deg
(D) 72 deg
(E) 36 deg
(F) 60 deg
(G) 30 deg

87. Which mirror brings a beam of parallel rays of light TOGETHER to a single point (a focus)? [Light]

(A) a flat glass sheet
(B) a concave mirror
(C) a plane mirror
(D) a convex mirror
(E) a polished flat metal plate
(F) none of these

88. A square's order of rotational symmetry and its number of lines of symmetry are added together. What is the SUM? [Symmetry]

(A) 4
(B) 6
(C) 16
(D) 12
(E) 8
(F) 2

89. A sunflower head has about 200 tiny florets, and each healthy floret makes one seed. If 35% of the florets stay empty, how many seeds does the head produce? [Reproduction in Plants]

- **(A)** 70 seeds
- **(B)** 130 seeds
- **(C)** 135 seeds
- **(D)** 200 seeds
- **(E)** 165 seeds
- **(F)** 65 seeds
- **(G)** 100 seeds

90. Can a triangle have sides 5 cm, 6 cm and 12 cm? Decide using the triangle inequality. [The Triangle & its Properties]

- **(A)** No, because 5 + 6 is less than 12
- **(B)** No, because the sides are too small
- **(C)** Yes, because all sides are different
- **(D)** Yes, because 5 + 12 is more than 6
- **(E)** Yes, because 6 + 12 is more than 5
- **(F)** Yes, because 12 is the longest side
- **(G)** No, because it must be right-angled

91. Two plane mirrors are placed at an angle of 45 deg. How many images are formed? [Light]

- **(A)** 4 images
- **(B)** 45 images
- **(C)** 6 images
- **(D)** 5 images
- **(E)** 8 images
- **(F)** 7 images
- **(G)** 9 images

92. A figure has rotational symmetry of order 6. After how many degrees of the SMALLEST turn does it look the same, and how many such positions are there in a full turn? [Symmetry]

- **(A)** 60 deg each turn, with 3 positions
- **(B)** 120 deg each turn, with 6 positions
- **(C)** 60 deg each turn, with 6 matching positions
- **(D)** 90 deg each turn, with 4 positions
- **(E)** 60 deg each turn, with 12 positions
- **(F)** 6 deg each turn, with 60 positions
- **(G)** 30 deg each turn, with 6 positions

93. Which statement about the reproductive parts of a flower is correct? [Reproduction in Plants]

- **(A)** the stigma makes pollen and the anther holds the egg
- **(B)** the filament makes pollen and the style holds the egg
- **(C)** the style makes pollen and the ovary makes pollen too
- **(D)** the ovary makes pollen and the anther holds the egg
- **(E)** the stigma makes the egg and the ovule makes pollen
- **(F)** the anther makes pollen (male) and the ovule contains the egg (female)

94. An equilateral triangle has a perimeter of 27 cm. What is the size of each of its angles, and the length of each side? [The Triangle & its Properties]

- **(A)** each angle 120 deg, each side 9 cm
- **(B)** each angle 60 deg, each side 13.5 cm
- **(C)** each angle 30 deg, each side 9 cm
- **(D)** each angle 45 deg, each side 9 cm
- **(E)** each angle 90 deg, each side 9 cm
- **(F)** each angle 60 deg, each side 27 cm
- **(G)** each angle 60 deg, each side 9 cm

95. Why is the word 'AMBULANCE' often painted reversed on the front of an ambulance? [Light]

- **(A)** to make the word travel faster
- **(B)** so it can be read from inside the ambulance
- **(C)** so the letters look pretty
- **(D)** so that drivers ahead read it correctly in their rear-view mirror (it undoes lateral inversion)
- **(E)** because red letters must be reversed
- **(F)** because mirrors make words larger
- **(G)** because paint dries that way

96. Which of these shapes has the GREATEST number of lines of symmetry? [Symmetry]

- **(A)** a rectangle (not a square)
- **(B)** a regular pentagon... no, fewer than this answer
- **(C)** a parallelogram
- **(D)** a regular hexagon
- **(E)** an isosceles triangle
- **(F)** a plain scalene triangle
- **(G)** a rhombus

97. Why do farmers often prefer vegetative propagation (cuttings, grafts) for fruit plants over growing them from seed? [Reproduction in Plants]

- **(A)** the new plants are always completely different from the parent
- **(B)** the new plants are genetically identical to the parent and flower or fruit sooner
- **(C)** it removes the need for any water
- **(D)** seeds can never grow into plants
- **(E)** it turns the plant into an animal
- **(F)** it stops the plant from ever making flowers

98. In a triangle, the largest angle is 84 deg and the smallest is 48 deg. Find the third angle, then state whether the triangle is acute, right or obtuse. [The Triangle & its Properties]

- **(A)** 54 deg, and the triangle is right-angled
- **(B)** 36 deg, and the triangle is obtuse-angled
- **(C)** 48 deg, and the triangle is right-angled
- **(D)** 48 deg, and the triangle is obtuse-angled
- **(E)** 52 deg, and the triangle is acute-angled
- **(F)** 48 deg, and the triangle is acute-angled

99. Sunlight reflects off a plane mirror onto a wall. If the angle of incidence is increased from 30 deg to 50 deg, what happens to the angle of reflection? [Light]

- **(A)** it stays at 30 deg
- **(B)** it also increases to 50 deg
- **(C)** it halves to 25 deg
- **(D)** it becomes 0 deg
- **(E)** it becomes 20 deg
- **(F)** it becomes 90 deg

100. A windmill is built with 4 identical curved blades, all swept the same way. It looks the same after a quarter turn. How many lines of symmetry does it have, and what is its rotational order? [Symmetry]

- **(A)** 1 line, order 4
- **(B)** 0 lines, order 2
- **(C)** 2 lines, order 4
- **(D)** 0 lines of symmetry, rotational order 4
- **(E)** 4 lines, order 4
- **(F)** 4 lines, order 1